

END TERM EXAMINATION**THIRD SEMESTER [B.TECH] DECEMBER 2024 – JANUARY 2025****Paper Code: CIC-211****Subject: Object-Oriented Programming Using C++****Time: 3 Hours****Maximum Marks: 60****Note:** Attempt five questions in all including Q. No. 1 which is compulsory. Select one question from each unit.

Q. No.	Question	Marks
1	Attempt any five from the following questions: (a) Explain the concept of return by reference. How to call it? Give an example. (b) Differentiate between Abstraction vs Encapsulation. (c) Explain the properties of static member functions. (d) 'Namespaces in C++ are used for organizing and avoiding naming conflicts in large projects.' Justify whether the given statement is True or False. Give suitable example. (e) Differentiate between Runtime vs Compile-time Polymorphism. (f) What are constructors? Can constructors be overloaded? Explain with an example.	4 × 5 = 20
UNIT – I		
2	(a) Develop a C++ program to represent a Bank Account. Create appropriate data members (account number, balance, name) and member functions (deposit, withdraw, display). (6 Marks) (b) Explain dynamic memory allocation in C++. Explain the use of new and delete operators. (4 Marks)	6 + 4 = 10
3	[OR] (a) Differentiate between: (3 + 3 = 6 Marks) (i) Implicit and explicit type conversion (ii) Call by Value and Call by Reference (b) Explain the concept of inline function. Where should it not be used? Develop a program to calculate the factorial of a number using an inline function. (4 Marks)	6 + 4 = 10
UNIT – II		
4	(a) What is an array of objects? How is it stored inside memory? Develop a program in C++ to manage a student database for 'n' students using an array of student objects. Each student should have attributes like name, ID, and marks. Implement member functions to input and display students' details. (6 Marks) (b) Describe the use of a constructor in C++. What are its characteristics? Write a program using different types of constructors in a class. Can constructors be inherited in a derived class? (4 Marks)	6 + 4 = 10
5	[OR] (a) Write short notes on the following with suitable examples: (2 + 2 = 4 Marks) (i) Use of destructor and its order of calling (ii) Dynamic memory allocation in C++ (b) Define operator overloading. Which operators in C++ cannot be overloaded? Implement a C++ class representing Time (hours, minutes, seconds). Overload the + operator to enable the addition of two time instances. (6 Marks)	4 + 6 = 10

[P.T.O.]

Q. No.	Question	Marks
UNIT – III		
6	(a) Define pure virtual functions and abstract classes. What are their properties? Provide an example illustrating their use. (4 Marks) (b) Develop a program in C++ to represent an employee hierarchy using inheritance. Create a base class Employee with attributes like name, ID, and salary. Derive classes for different types of employees, such as Manager and Technician, each with specific attributes. Include member functions to display details of each employee type. (6 Marks) <pre> graph TD Employee[Employee] --> Manager[Manager] Employee --> Technician[Technician] </pre>	4 + 6 = 10
7	[OR] (a) Write short notes on the following with example: (3 + 3 = 6 Marks) (i) Function overriding (ii) Aggregation vs Composition (b) What is generic programming? Design a generic C++ template function that swaps the values of two variables. Write a program that uses this template function to swap integers, doubles, and characters. Ensure that your template function works for various data types. (4 Marks)	6 + 4 = 10
UNIT – IV		
8	(a) Differentiate between the following: (4 × 1.5 = 6 Marks) (i) Iterators and pointers (ii) Queue and deque (iii) Arrays and vectors (iv) Sets and maps (b) What is the Standard Template Library (STL)? What are its various components? What is the purpose of iterators in an STL container? Explain the concept of vector iterators using the begin() and end() functions. (4 Marks)	6 + 4 = 10
9	[OR] Write short notes on any two of the following: (5 + 5 = 10 Marks) (a) Persistent objects (b) Exception handling in C++ (c) Vectors in STL	5 + 5 = 10
